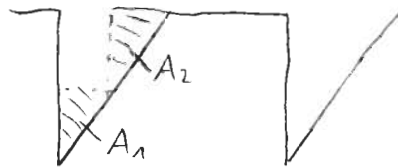


1 Aufgabe

$$\begin{aligned}
 1a) \quad \bar{u} &= \frac{1}{T} \int_0^T u(t) dt \\
 &= \frac{1}{T} \left[\int_0^{aT} u(t) dt + \int_{aT}^T u(t) dt \right] \\
 &= \frac{1}{T} \left[\int_0^{aT} \left(-U_1 + \frac{2U_1}{aT} t \right) dt + \int_{aT}^T U_1 dt \right] \\
 &= \frac{1}{T} \left[-U_1 t + \frac{2U_1}{aT} \frac{t^2}{2} \right]_0^{aT} + U_1 t \Big|_{aT}^T \\
 &= \frac{1}{T} \left[-U_1 aT + \frac{U_1}{aT} (aT)^2 + U_1 (T - aT) \right] \\
 &= \frac{1}{T} \left[-U_1 aT + U_1 aT + U_1 T (1-a) \right] \\
 &= U_1 (1-a)
 \end{aligned}$$

anderer Weg :

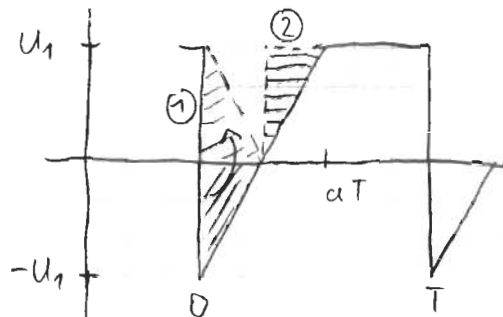


$A_1 = A_2$ haben sich auf

$$\Rightarrow \bar{u} = \frac{1}{T} \int_{aT}^T U_1 dt = \frac{U_1}{T} (T - aT)$$

$$= \underline{\underline{U_1 (1-a)}}$$

$$1b) \quad \overline{|u|} = \frac{1}{T} \int_0^T |u| dt$$



① Untere Dreiecksfläche hochklappen

② Verschieben und Drehen

$$\begin{aligned} \Rightarrow |\bar{u}| &= \frac{1}{T} \int_{\frac{a}{2}T}^T u_1 dt \\ &= \frac{1}{T} u_1 \left(T - \frac{a}{2} T \right) \\ &= \underline{\underline{u_1 \left(1 - \frac{a}{2} \right)}} \end{aligned}$$

$$1c) \quad u_{\text{eff}} = \sqrt{\frac{1}{T} \int_0^T u^2(t) dt}$$

$$= \left\{ \frac{1}{T} \left[\int_0^{aT} \left(-u_1 + \frac{2u_1}{aT} t \right)^2 dt + \int_{aT}^T u_1^2 dt \right] \right\}^{-\frac{1}{2}}$$

$$= \left\{ \frac{1}{T} \left[\int_0^{aT} \left(u_1^2 - \frac{4u_1^2}{aT} t + \frac{4u_1^2}{a^2 T^2} t^2 \right) dt + u_1^2 (T - aT) \right] \right\}^{-\frac{1}{2}}$$

$$= \left\{ \frac{1}{T} \left[u_1^2 aT - \frac{4u_1^2}{aT} \frac{(aT)^2}{2} + \frac{4u_1^2}{a^2 T^2} \frac{(aT)^3}{3} + u_1^2 (T - aT) \right] \right\}^{-\frac{1}{2}}$$

$$= \left\{ \frac{1}{T} \left[\cancel{u_1^2 aT} - 2u_1^2 aT + \frac{4}{3} u_1^2 aT + u_1^2 T - \cancel{u_1^2 aT} \right] \right\}^{-\frac{1}{2}}$$

$$= \left\{ \frac{1}{T} \left[-\frac{2}{3} aT u_1^2 + u_1^2 T \right] \right\}^{-\frac{1}{2}}$$

$$= \left(-\frac{2}{3} \alpha U_1^2 + U_1^2 \right)^{-\frac{1}{2}}$$

$$= U_1 \sqrt{1 - \frac{2}{3} \alpha}$$

$$1d) \quad P = R \cdot I_{\text{eff}}^2 = \frac{U_{\text{eff}}^2}{R} \stackrel{!}{=} \frac{(5,5 \text{ V})^2}{R}$$

$$\Rightarrow U_{\text{eff}} \stackrel{!}{=} 5,5 \text{ V} = U_1 \sqrt{1 - \frac{2}{3} \alpha} \quad (\text{Gl 1})$$

$$\overline{U} \stackrel{!}{=} 5,0 \text{ V} = U_1 \left(1 - \frac{\alpha}{2} \right) \quad (\text{Gl 2})$$

Gl 1 / Gl 2

$$\Rightarrow \frac{5,5 \text{ V}}{5,0 \text{ V}} = \frac{\sqrt{1 - \frac{2}{3} \alpha}}{1 - \frac{\alpha}{2}}$$

$$1,1^2 = \frac{(1 - \frac{2}{3} \alpha)}{(1 - \frac{\alpha}{2})^2}$$

$$1,1^2 \left(1 - \frac{\alpha}{2} \right)^2 = 1 - \frac{2}{3} \alpha$$

$$1,1^2 \left(1 - \alpha + \frac{\alpha^2}{4} \right) = 1 - \frac{2}{3} \alpha$$

$$1,21 - 1,21 \alpha + 0,3025 \alpha^2 - 1 + \frac{2}{3} \alpha = 0$$

$$0,21 - 0,543 \alpha + 0,3025 \alpha^2 = 0$$

$$\alpha^2 - 1,796 \alpha + 0,694 = 0$$

$$\alpha_{1,2} = + \frac{1,796}{2} \pm \frac{1}{2} \sqrt{1,796^2 - 4 \cdot 0,694}$$

$$= 0,898 \pm \frac{1}{2} 0,671$$

$$\alpha_1 = 1,233 \quad \text{nicht möglich}$$

$$\alpha_2 = \underline{\underline{0,563}}$$

aus (Gl. 2)

$$U_1 = \frac{5 \text{ V}}{1 - \frac{\alpha}{2}} = \underline{\underline{6,96 \text{ V}}}$$

2. Aufgabe

$$2a) \underline{U}_A = 5V \cdot e^{-j90^\circ} = \underline{\underline{-j5V}}$$

$$\begin{aligned} b) \underline{Z}_{C1} = \underline{Z}_{C2} &= \frac{1}{j\omega C_1} = \frac{1}{j2\pi f \cdot C_1} \\ &= \frac{1}{j2\pi \cdot 1591,55 \cdot \frac{1}{5} \cdot 80\mu F} \\ &= \underline{\underline{-j1,25\Omega}} \end{aligned}$$

$$\begin{aligned} \underline{Z}_L &= j\omega L = j2\pi f \cdot L \\ &= j2\pi \cdot 1591,55 \cdot \frac{1}{5} \cdot 0,125\text{ mH} \\ &= \underline{\underline{j1,25\Omega}} \end{aligned}$$

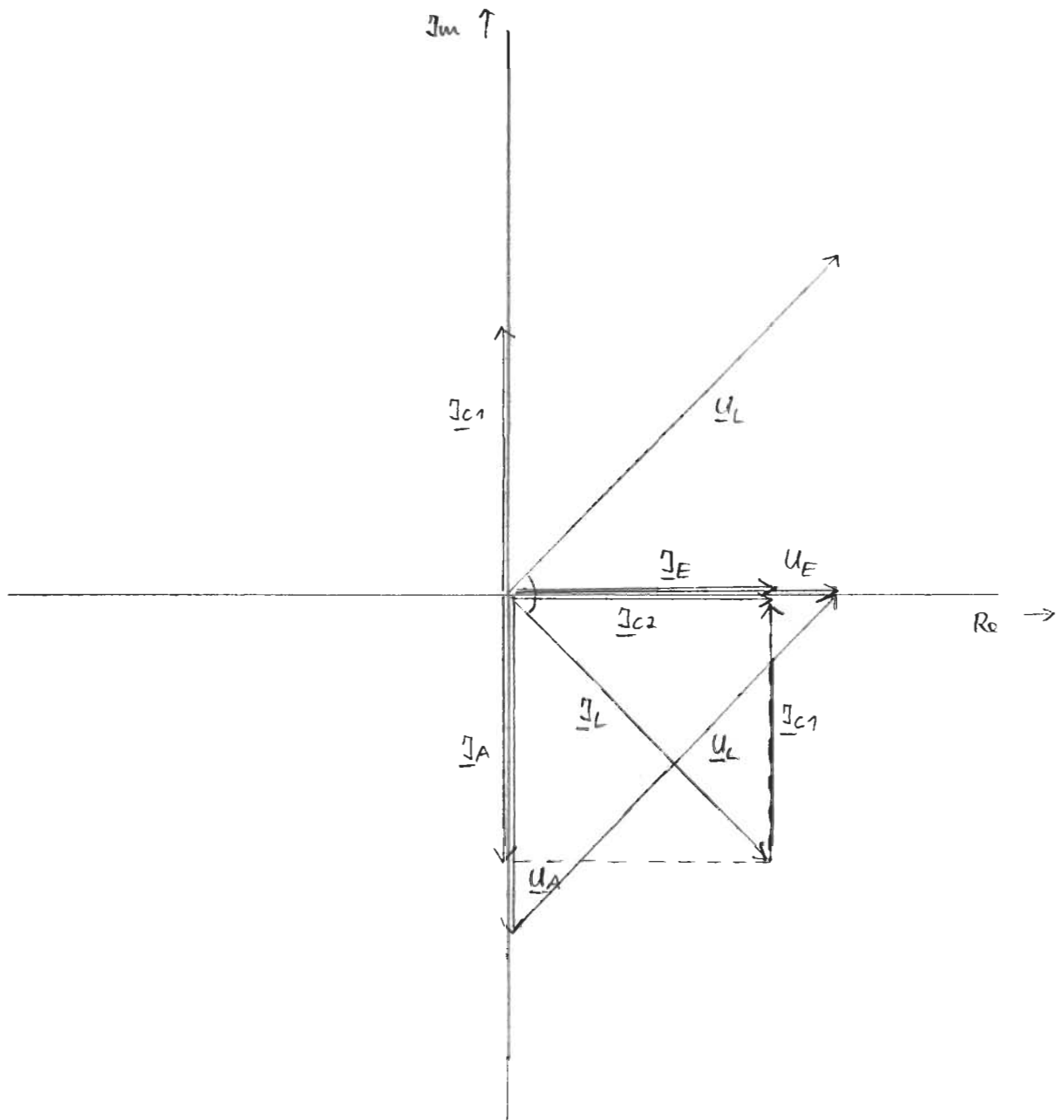
$$c) \underline{I}_A = \frac{\underline{U}_A}{R} = \frac{-j5V}{1,25\Omega} \cdot \underline{\underline{-j4A}} = 4A e^{-j90^\circ}$$

$$d) \underline{I}_{C2} = \frac{\underline{U}_A}{\frac{1}{j\omega C_1}} = \frac{\underline{U}_A}{\underline{Z}_C} = \frac{-j5V}{-j1,25\Omega} = \underline{\underline{4A}}$$

$$\begin{aligned} e) \underline{I}_L &= \underline{I}_A + \underline{I}_{C2} \\ &= -j4A + 4A = \underline{\underline{4A(1-j)}} \\ &= \sqrt{2} \cdot 4A e^{-j45^\circ} = 5,66V e^{-j45^\circ} \end{aligned}$$

$$\begin{aligned} \Rightarrow i_L(t) &= \sqrt{2} \cdot (\sqrt{2} \cdot 4A) \cos(\omega t - 45^\circ) \\ &= \underline{\underline{8A \cos(\omega t - 45^\circ)}} \end{aligned}$$

$$\begin{aligned} f) \underline{U}_L &= \underline{Z}_L \underline{I}_L \\ &= j1,25\Omega \cdot \sqrt{2} \cdot 4A e^{-j45^\circ} \\ &= \sqrt{2} \cdot 5V e^{j45^\circ} = \underline{\underline{7,07V e^{j45^\circ}}} \end{aligned}$$



$$\begin{aligned}
 g) \quad \underline{U}_E &= \underline{U}_L + \underline{U}_A \\
 &= 7,07 \text{ V } e^{j45^\circ} - j5 \text{ V} \\
 &= 7,07 \text{ V } \cos 45^\circ + j 7,07 \sin 45^\circ - j5 \text{ V} \\
 &= 5 \text{ V} + j5 \text{ V} - j5 \text{ V} \\
 &= \underline{\underline{5 \text{ V}}}
 \end{aligned}$$

(oder Ablesen aus Diagramm $5 \text{ cm} \Rightarrow U_E = 5 \text{ V}$)

$$\angle(\underline{U}_A, \underline{U}_E) = -90^\circ$$

Die Ausgangsspannung eilt der Eingangsspannung um 90° nach.

$$h) \quad \underline{I}_{C1} = \frac{\underline{U}_E}{\underline{Z}_{C1}} = \frac{5 \text{ V}}{-j1,25 \Omega} = j4 \text{ A}$$

$$\begin{aligned}
 \underline{I}_E &= \underline{I}_{C1} + \underline{I}_L = j4 \text{ A} + 4 \text{ A} (1 - j) \\
 &= \underline{\underline{4 \text{ A}}}
 \end{aligned}$$

$$\begin{aligned}
 i) \quad P_R &= R \cdot I_A^2 = 1,25 \Omega \cdot (4 \text{ A})^2 \\
 &= 20 \text{ W}
 \end{aligned}$$

$$P_{U_E} = P_R = 20 \text{ W}$$

$$\begin{aligned}
 j) \quad \underline{U}_E \text{ und } \underline{I}_E \text{ in Phase} &\Rightarrow \text{nur Wirkleistung} \\
 \Rightarrow \underline{S} = P_R &= 20 \text{ VA}, \quad Q = 0 \text{ VAR}
 \end{aligned}$$

k) Quelle nicht nur ohmsche Last
 $\Rightarrow$ Resonanz

$$3a) \quad \underline{A}_1(\omega) = \frac{\frac{1}{j\omega C}}{R_1 + \frac{1}{j\omega C}} = \frac{1}{1 + j\omega R_1 C}$$

$$\underline{A}_2(\omega) = 1$$

$$\underline{A}_3(\omega) = \frac{R_2}{R_2 + j\omega L} = \frac{1}{1 + j\omega \frac{L}{R_2}}$$

$$b) \quad T_1 = R_1 C = 1k\Omega \cdot 79nF = 79\mu s$$

$$f_{g1} = \frac{1}{2\pi T_1} = 2015 \text{ Hz} \approx 2000 \text{ Hz}$$

$$T_3 = \frac{L}{R_2} = \frac{31,8mH}{1k\Omega} = 31,8\mu s$$

$$f_{g3} = \frac{1}{2\pi T_3} = 5005 \text{ Hz} \approx 5000 \text{ Hz}$$

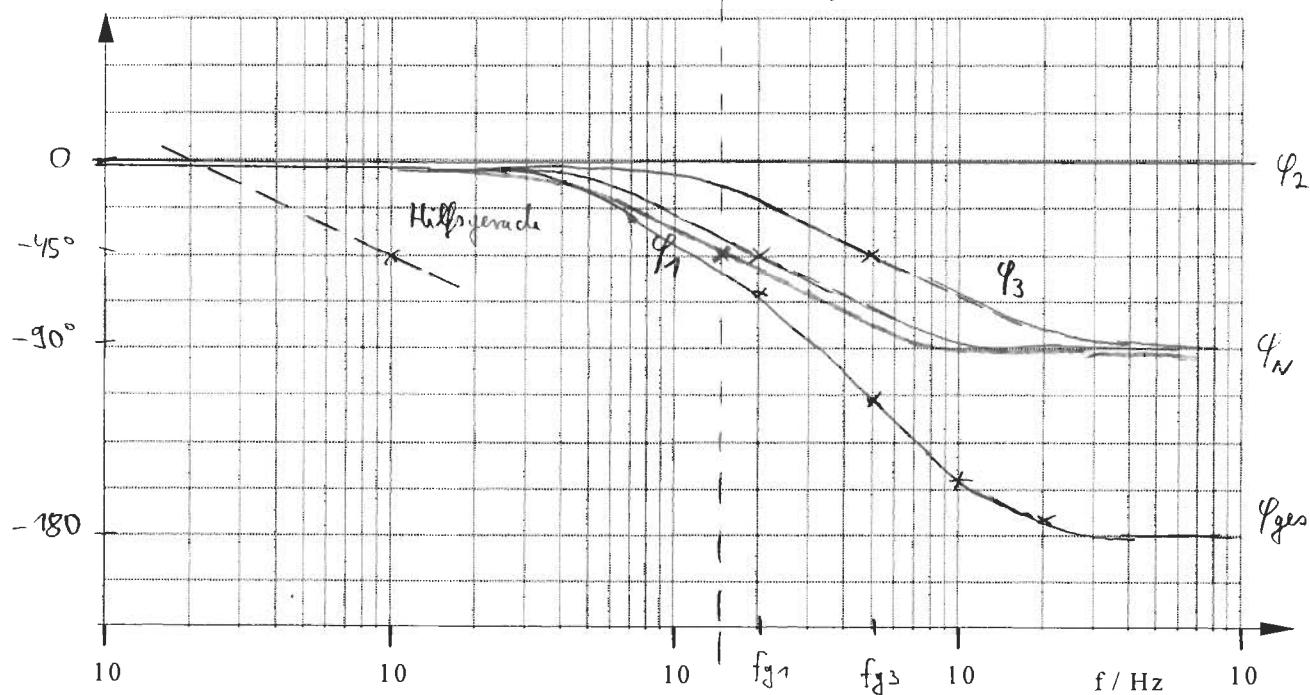
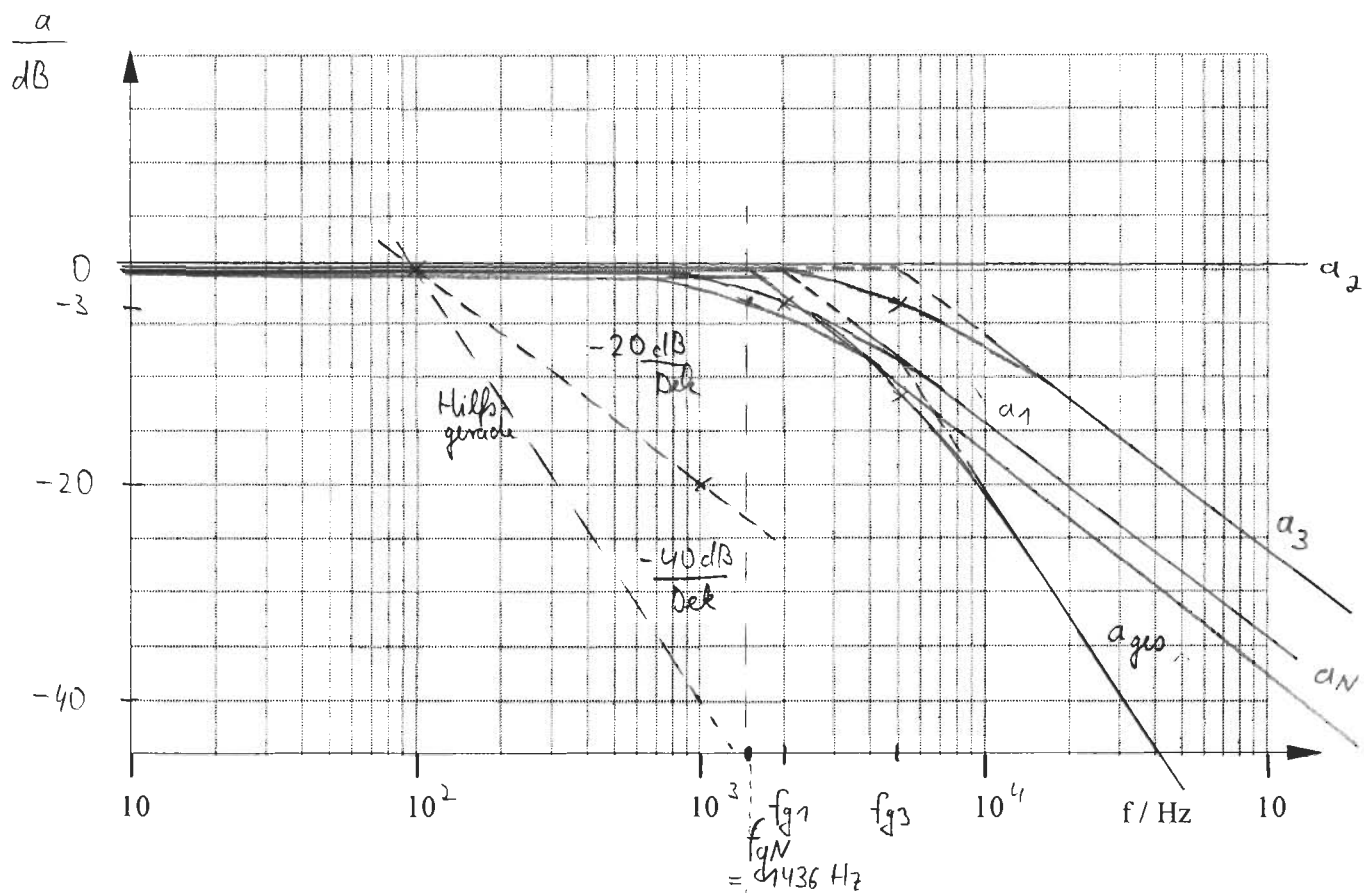
c) siehe Diagramm

$$d) \quad \underline{A}(f) = \frac{1}{1 + j \frac{f}{f_{g1}}} \cdot 1 \cdot \frac{1}{1 + j \frac{f}{f_{g3}}}$$

$$e) \quad \underline{A}_N(f) = \frac{1}{1 + j \frac{f}{f_{gN}}}$$

$$\text{mit } f_{gN} = \frac{1}{2\pi (T_1 + T_2)} = \frac{1}{2\pi (79\mu s + 31,8\mu s)} \\ = 1436 \text{ Hz}$$

Bodediagramm:



g) Amplitudengang $\begin{cases} a_N & \text{fällt mit } -20 \text{ dB/Dek} \\ a_{ges} & \text{" " } -40 \text{ "} \end{cases}$

Phasengang $\begin{cases} \varphi_N & \text{geht auf } -90^\circ \\ \varphi_{ges} & \text{" " } -180^\circ \end{cases}$

h) $|A| = \frac{1}{\sqrt{1 + \left(\frac{f}{f_{g1}}\right)^2}} \cdot \frac{1}{\sqrt{1 + \left(\frac{f}{f_{g3}}\right)^2}}$

$$= \frac{1}{\sqrt{\left(1 + \left(\frac{f}{f_{g1}}\right)^2\right) \left(1 + \left(\frac{f}{f_{g3}}\right)^2\right)}}$$

$$|A_N| = \frac{1}{\sqrt{1 + \left(\frac{f}{f_{gN}}\right)^2}}$$

$$\Rightarrow \left[1 + \left(\frac{f_0}{f_{g1}}\right)^2\right] \left[1 + \left(\frac{f_0}{f_{g3}}\right)^2\right] = \left[1 + \left(\frac{f_0}{f_{gN}}\right)^2\right]$$

$$\Leftrightarrow 1 + \left(\frac{f_0}{f_{g1}}\right)^2 + \left(\frac{f_0}{f_{g3}}\right)^2 + \left(\frac{f_0^2}{f_{g1} f_{g3}}\right)^2 = \left(\frac{f_0}{f_{gN}}\right)^2 + 1$$

$$\Leftrightarrow \left(\frac{f_0}{f_{g1}}\right)^2 + \left(\frac{f_0}{f_{g3}}\right)^2 + \left(\frac{f_0^2}{f_{g1} f_{g3}}\right)^2 = \left(\frac{f_0}{f_{gN}}\right)^2$$

$$f_0^2 \left[\frac{1}{f_{g1}^2} + \frac{1}{f_{g3}^2} + \frac{f_0^2}{f_{g1}^2 f_{g3}^2} - \frac{1}{f_{gN}^2} \right] = 0$$

$$\Rightarrow \frac{f_0^2}{f_{g1}^2 f_{g3}^2} = \frac{1}{f_{gN}^2} - \frac{1}{f_{g1}^2} - \frac{1}{f_{g3}^2}$$

$$f_0^2 = \frac{f_{g1}^2 f_{g3}^2}{f_{gN}^2} - f_{g3}^2 - f_{g1}^2$$

$$f_0^2 = \left(\frac{f_{g1} f_{g3}}{f_{gN}}\right)^2 - f_{g3}^2 - f_{g1}^2$$

$$f_0 = \sqrt{\left(\frac{f_{g1} f_{g3}}{f_{gN}}\right)^2 - f_{g3}^2 - f_{g1}^2}$$

$$f_0 = \sqrt{\left(\frac{2015 \cdot 5005}{1436}\right)^2 - 2015^2 - 5005^2} \text{ Hz}$$

$$= 4496 \text{ Hz}$$

$$i) |A_N(f_0)| = \frac{1}{\sqrt{1 + \left(\frac{4496}{1436}\right)^2}} = 0,304 \quad (\hat{=} -10,3 \text{ dB})$$

$$j) \varphi_N(f_0) = -\arctan\left(\frac{f_0}{f_{gN}}\right)$$

$$= -\arctan\left(\frac{4496 \text{ Hz}}{1436 \text{ Hz}}\right)$$

$$= -72,3^\circ$$

$$\varphi(f_0) = -\arctan\left(\frac{f_0}{f_{g1}}\right) - \arctan\left(\frac{f_0}{f_{g3}}\right)$$

$$= -\arctan\left(\frac{4496 \text{ Hz}}{2015 \text{ Hz}}\right) - \arctan\left(\frac{4496 \text{ Hz}}{5005 \text{ Hz}}\right)$$

$$= -107,8^\circ$$